

CO-1
Assessment Tool-1

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course: C-program.

- 1) The company wants to automate its payroll process.
Write a C program to calculate the total Salary by adding the Basic salary & Bonus amount of an employee.

A) pseudo code :-

Start

Read Basic Salary, Bonus

Total Salary = Basic salary + Bonus

print total salary

stop

Input :-

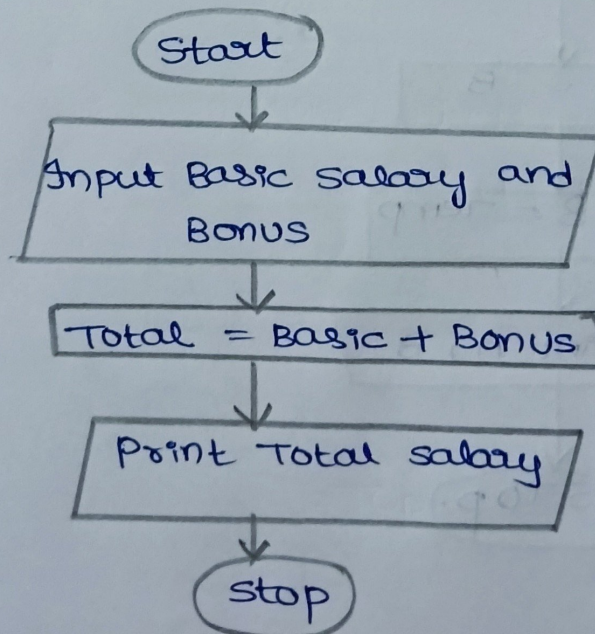
Basic salary

Bonus

output :-

Total Salary.

Flow chart :-



2) During a database update, two employee IDs need to be exchanged. Write a C-program to swap two numbers using a temporary variable.

A) pseudo code :-

Start

Read A, B

Temp = A

A = B

B = Temp

Print A, B

Stop

Input :-

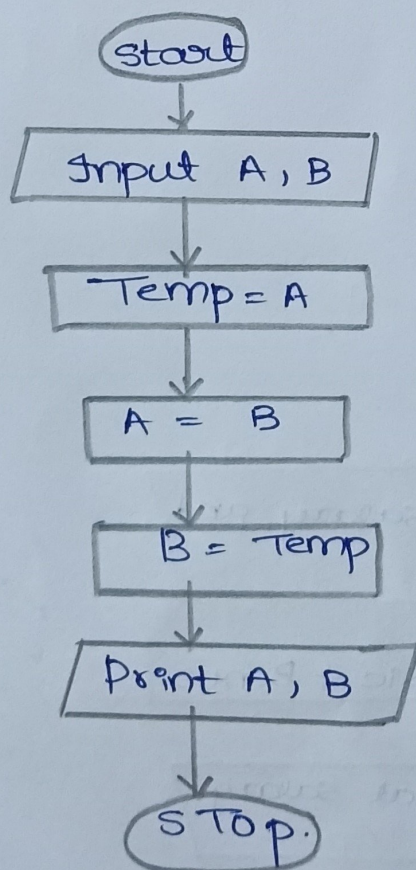
A

B

output :-

Swapped values of A and B.

Flowchart :-



3) In an embedded system with limited memory two values must be exchanged without using any extra storage. write a c-program to swap two numbers without using Temporary variable.

A) pseudo code :-

Start
Read A, B
 $A = A + B$
 $B = A - B$
 $A = A - B$
Print A, B
STOP

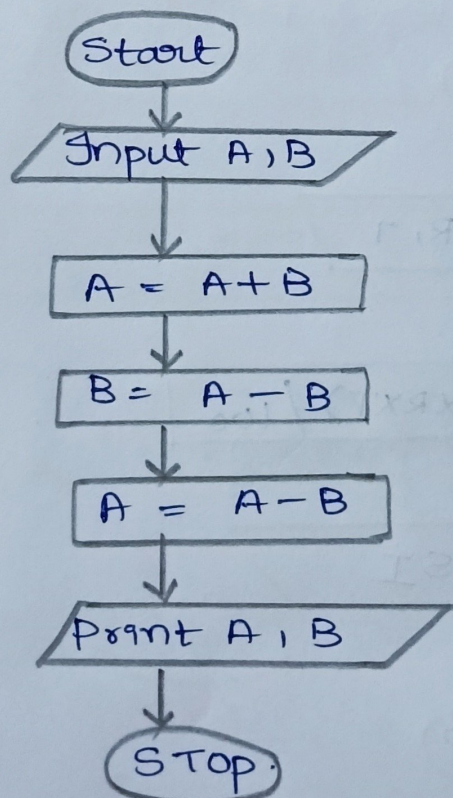
Input :-

A
B

output :-

Swapped values of A and B

flowchart :-



- 4) A bank wants to provide customers with an estimate of the interest earned on their deposit write a C program to calculate simple interest using the formula.

A) pseudo code :-

Start

Read principal, Rate, Time

$SI = (\text{principal} \times \text{Rate} \times \text{Time}) / 100$

print SI

Stop

Input :-

Principal (P)

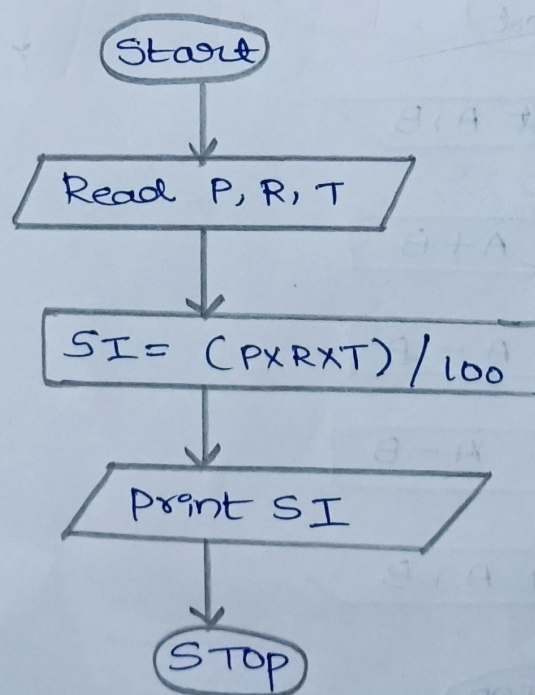
Rate (R)

Time (T)

output :-

Simple interest

flow chart :-



- 5) A weather station records temperature in celsius but reports need to generate in Fahrenheit. write a c program to convert temperature from celsius to fahrenheit.

pseudo code :-

Start

Read celsius

Fahrenheit = $(\text{celsius} \times 9/5) + 32$

print fahrenheit

Stop

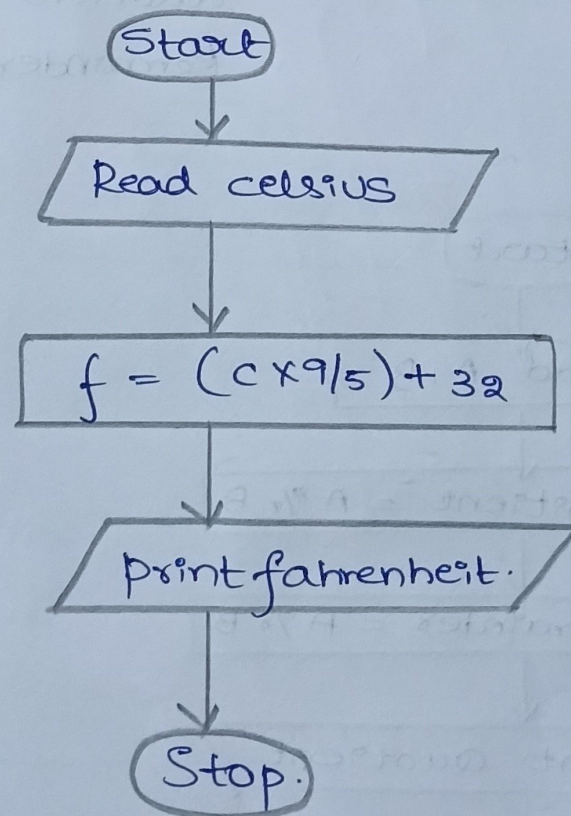
Input :-

Celsius

output :-

fahrenheit

flowchart :-



- 6) A mathematics learning application need to display both the quotient & remainder. When two numbers are divided. Write a C program to find the Quotient and remainder of two integers.

A) pseudo code :-

Start

Read A, B

Quotient = A/B

remainder = $A \% B$

print Quotient

print remainder

Stop.

Input :-

Dividend (A)

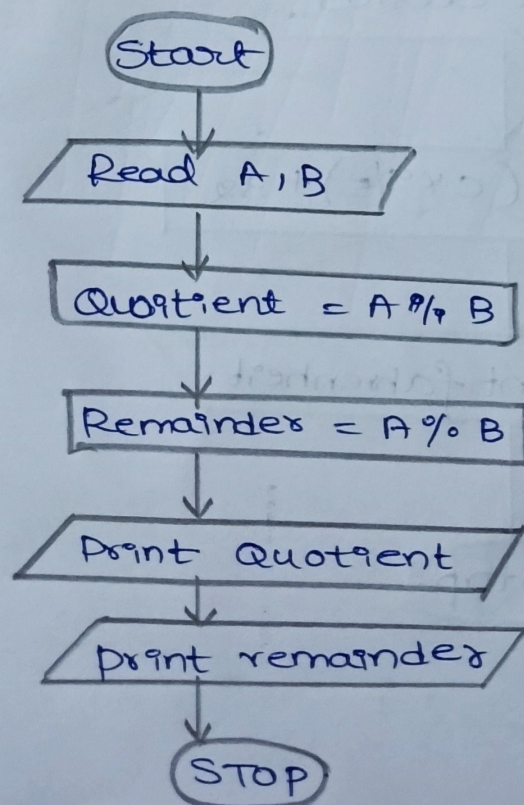
Divisor (B)

output :-

Quotient

Remainder.

flowchart :-



7) An event management system allows entry based on even-numbered ticket IDs. Write a C program to determine whether a given ticket number is even or odd.

A) pseudo code :-

Start

Read N

If $N \% 2 == 0$

Print "even"

else

print "odd"

Stop

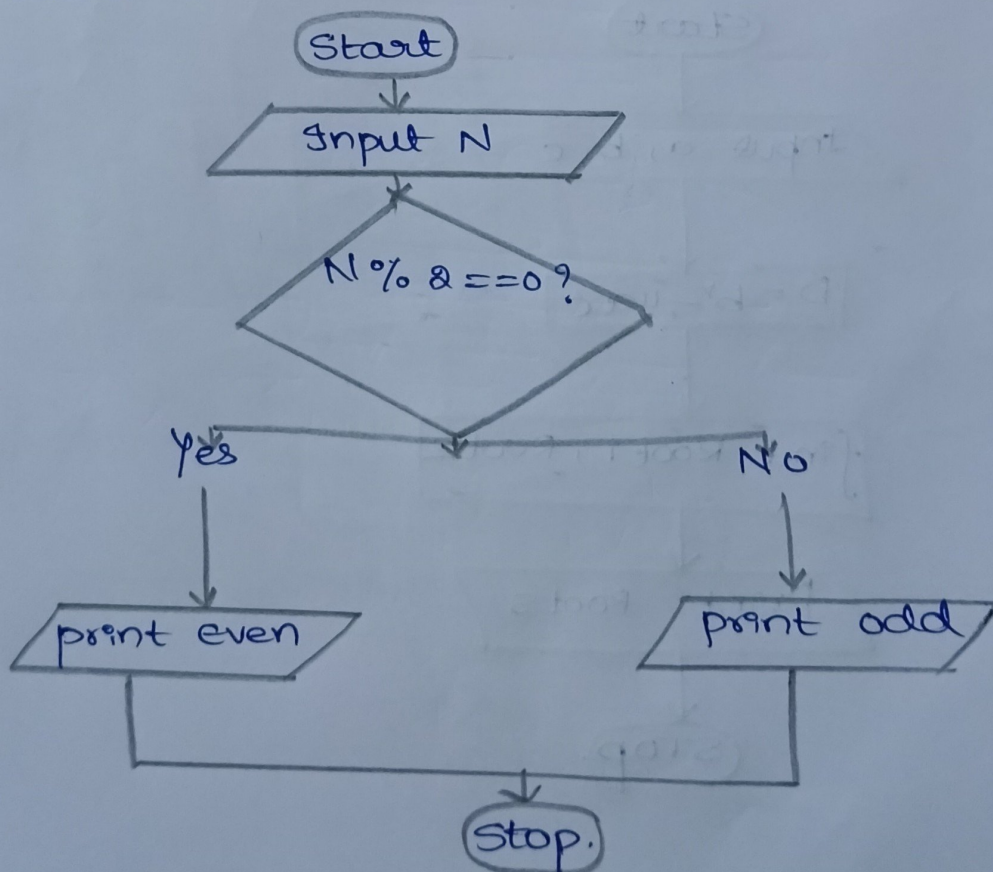
Input

integer N

output

Even or odd.

flow chart :-



8) An engineer needs to solve quadratic equation of the form. write a program to calculate and display the roots of the quadratic equation.

A) pseudo code:-

Start

Read a, b, c

$$D = b^2 - 4 * a * c$$

$$\text{Root 1} = (-b + \sqrt{D}) / (2 * a)$$

$$\text{Root 2} = (-b - \sqrt{D}) / (2 * a)$$

print Root 1, Root 2

Stop

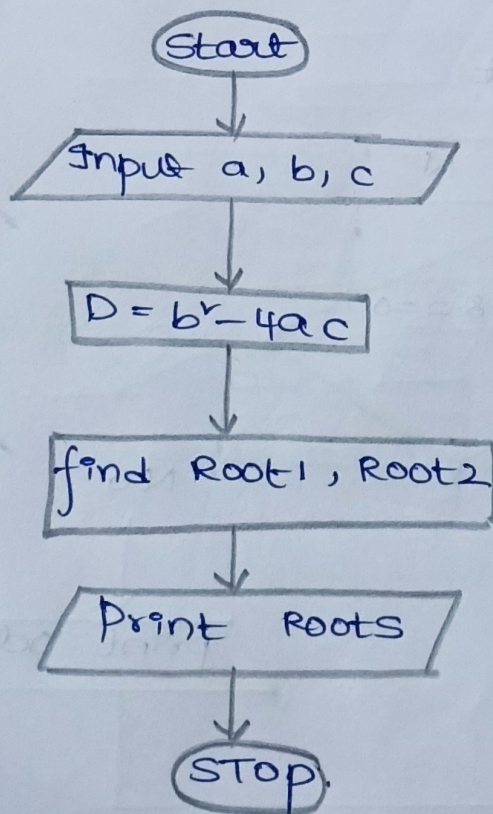
Input:-

a, b, c

output:-

Root 1 and Root 2.

flowchart:-



- 9) A scientific calculator application requires the square & cube of a given number. write a program to calculate the square and cube of a number entered by the user.

A) pseudo code :-

Start

Read N

Square = $N * N$

Cube = $N * N * N$

print Square, cube

Stop

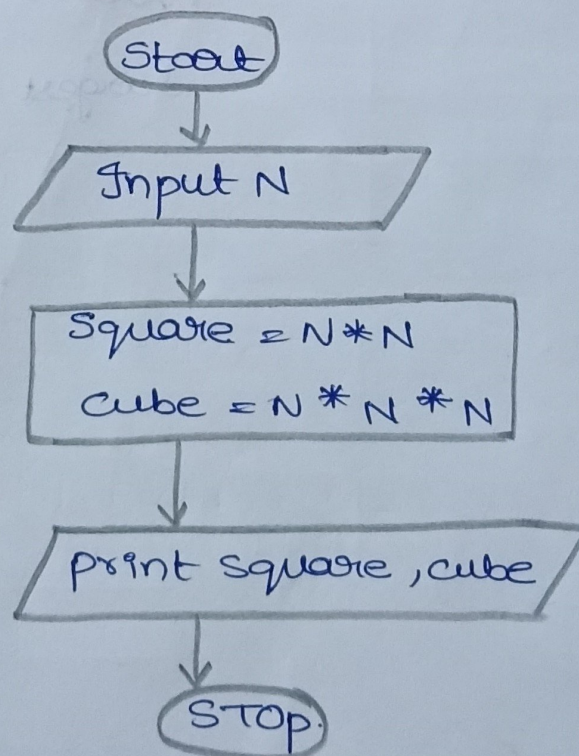
Input :-

Number N

output :-

Square & cube

flow chart :-



10) A company wants to identify the top-performing salesperson among three employees based on their sales figures. Write a C program to find the largest among the three numbers.

A) Input pseudo code :-

```
Start
Read A, B, C
If  $A > B$  and  $A > C$ 
    print A
else if  $B > A$  and  $B > C$ 
    print B
else
    print C
Stop.
```

Input :-

A, B, C

Output :-

largest number.

Flowchart :-

